ForgeX

PSU_G0P20N Technical Reference

Project: ForgeX
Generation: Gen0
Document: PSU_G0P20N Technical Reference
Author: Omar Magdy
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Table of Contents

1. Introduction

1.1 Naming

`PSU_G0P20N` follows the ForgeX module naming convention:

1.2 Overview

2. Interfaces & I/O

2.1 Power Interface

2.2 Low-Voltage Interface

2.3 Non Isolated Tie Interface

3. Design Requirements

- 400 V DC-link operation

4. Constraints

The Gen0 HBM design is subject to several practical constraints.

A significant constraint is the limited local supply chain, with component availability primarily governed by vendors accessible in Egypt. Component selection therefore considers availability and replacement options in addition to electrical performance.

PCB fabrication is constrained by the DFM rules applicable to the available fabrication process:

- Maximum number of layers: 2
- Minimum track width: 0.25 mm
- Minimum clearance: 0.25 mm
- Minimum polygon-pour clearance: 0.3 mm
- Minimum via diameter: 0.9 mm
- Minimum via hole diameter: 0.4 mm
- Minimum annular ring: 0.5 mm
- Maximum single-board size: 38 cm × 28 cm
- Maximum double-board size: 28 cm × 22 cm

The Gen0 implementation additionally prioritizes:

- Cost effectiveness
- Component availability
- Simplicity
- Reliability
- Ease of debugging and modification
- Maintainability during development

5. Sizing and Component Selection

5.1 Off-line Switcher

5.2 Flyback transformer

5.3 RCD Clamp

5.4 Feedback & Compensation

6. Layout Considerations & Highlights

7. Design Files

The complete hardware design files for this module are maintained in the ForgeX repository:

revA:

- PCB & Schematic
- Simulation
- **Manufacturing files:** 🛠

8. Simulation

Simulation is used as a first-pass validation and design tool for the HBM.

The simulations are intended to:

- Validate initial electrical estimates
- Evaluate the effects of layout-related parasitics
- Assist with component sizing
- Investigate switching behavior and transient effects
- Provide a basis for first-pass design tuning

Simulation results are not considered a substitute for physical testing and validation. Their accuracy depends on the quality, fidelity, and applicability of the underlying semiconductor, parasitic, and system models.

The primary simulation focus is the power-electronics behavior of the module and its associated switching infrastructure.

Switching Model

Small Signal Model

8.5 Digest and Conclusion

9. Field Tests and Validation 🛠

[Document laboratory testing, measurements, test conditions, and comparison against simulation.]

10. Known Issues and Limitations 🛠

[Document known limitations of Rev. A / Gen0.]

11. Revisions

Revision	Date	Description
Rev. 0	August 2026	Initial technical reference